

Multiple Choice (10 marks)

1. $\frac{2e^{-4s}}{s^3}$

- (a) $\frac{2}{s^3}$
- (b) $(\frac{2}{s^2}) * e^{-4s}$
- (c) $\frac{4}{s^3}$

$\frac{2e^{-4s}}{s^4}$

2. The maximum effective aperture of this dipole.

- (a) $0.075 \lambda^2$
- (b) $0.200 \lambda^2$
- (c) $0.180 \lambda^2$
- (d) $0.089 \lambda^2$
- (e) $0.102 \lambda^2$

3. $n_{11} = 0.25 \text{ mho}$, $y_{12} = -0.05 \text{ mho}$, $y_{22} = 0.1 \text{ mho}$. Find the z-parameters.

- (a) $z_{11} = 11.1 \Omega$, $z_{12} = z_{21} = 4.44 \Omega$, $z_{22} = 22.2 \Omega$
- (b) $z_{11} = 11.1 \Omega$, $z_{12} = z_{21} = 2.22 \Omega$, $z_{22} = 4.44 \Omega$
- (c) $z_{11} = 2.22 \Omega$, $z_{12} = z_{21} = 4.44 \Omega$, $z_{22} = 11.1 \Omega$
- (d) $z_{11} = 4.44 \Omega$, $z_{12} = z_{21} = 1.11 \Omega$, $z_{22} = 22.2 \Omega$
- (e) $z_{11} = 4.44 \Omega$, $z_{12} = z_{21} = 2.22 \Omega$, $z_{22} = 22.2 \Omega$

4. The maximum effective aperture of this dipole.

- (a) 5 percent
- (b) 40 percent
- (c) 35 percent
- (d) 10 percent
- (e) 20 percent

5. $\frac{AB}{|76| |75| |168|}$

- (a) $AB = |76| |75| |168|$
- (b) $AB = |96| |65| |158|$

- (c) $AB = |76| |45| |138|$
 (d) $AB = |96| |55| |138|$
 (e) $AB = |86| |45| |138|$

True / False (10 marks)

Answer True or False

6. $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$
7. $\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1$
8. $\lim_{x \rightarrow 0} \frac{\ln(1+x)}{x} = 1$
9. $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \frac{1}{2}$
10. $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Find the limit $\lim_{x \rightarrow 0} \frac{\sin x}{x}$ using L'Hôpital's rule.
12. Let $F = kr^n \hat{s}$ over a sphere of radius a centered at the origin. Discuss the implications of your results.

End of Paper